MIL-P-24289(SHIPS)
23 March 1967

MILITARY SPECIFICATION

PRESSING EQUIPMENT, TROUSER,

AUTOMATIC, ROTATING

NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers automatic rotating equipment consisting of two components, one rotating component consisting of ten trouser topping presses and one rotating component consisting of five trouser legging presses, for pressing trousers aboard Naval ships.

2. APPLICABLE DOCUMENTS

2.1 Not applicable to this specification.

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification .

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

Standard 45 - Recommended Practice for Electric Installations on Shipboard.

FSC 3510

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(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, Box A, Lenox Hill Station, New York, New York 10021.)

AMERICAN STANDARDS ASSOCIATION, INC.

ASA-A13a-1928 - Scheme for the Identification of Piping Systems.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016).

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

3.2 Material. - Materials used shall be equal to or better than the quality used by the manufacturer in his regular commercial production. No wood shall be used in the construction of the equipment. Operators' platforms and supporting structure shall be steel.

3.3 Quality. - The design, materials, construction, fabrication, finish, and operation of the equipment shall be suitable and adequate for the purpose intended. All surfaces of the equipment shall be finished in such a manner as not to injure clothing being pressed or operating personnel.

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3.4 Completeness. - Each unit shall be new and furnished complete with required structure, operating tracks, operating platforms, guard rails, presses, motors, driving mechanisms, safety mechanisms, starters, controllers, timers, piping, and all other equipments and appliances, as applicable, to efficiently perform the purpose for which intended when installed on a ship's deck and connected to the ship's power, steam and air systems.

3.5 Design. - Each unit shall essentially consist of trouser topping and legging presses mounted on cylindrical bases that rotate at predetermined speeds, whereby the operator of each component, from his operating platform, can lay garments on and remove garments from the presses as they rotate past his position. The press heads shall close automatically after passing the operating position and shall open automatically prior to reaching the operating position.

3.5.1 Arrangement. - Each automatic rotating trouser pressing unit shall consist of two cylindrical rotating frames or bases for supporting trouser topping presses and double trouser legging presses. The two rotating cylindrical frames or bases shall be designed for mounting side by side at an approximate distance between centers of 13 feet 6 inches.

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The outside diameter of the double trouser legging component shall be approximately 13 feet 6 inches and shall be fitted with five double trouser legging presses spaced equally around the circumference with fronts of presses facing out. The outside diameter of the trouser topping component shall be approximately 11 feet 6 inches and shall be fitted with ten trouser topping presses spaced equally around the circumference with fronts of presses facing out. A stationary operator's platform, at suitable operating height, shall be provided for each component. Each platform is to extend outward from the edge of the cylinder (but clear of cylinder to permit cylinder rotation) for approximately 2 feet 6 inches and extend around each cylinder for a distance equivalent to 60 to 65 degrees. The components shall be so arranged that the operators' platforms abutt, the right end of the platform for the topping component abutting the left end of the platform for the legging component.

3.5.2 Speed and direction of rotation. - Both components, topping and legging, shall rotate from left to right, or counter clockwise. Speed of rotation shall be variable, at discretion of operator, above and below design speed. Design speed shall be one revolution each 45 seconds for the topping component and one revolution each 90 seconds for the legging component to thoroughly and neatly press 200 trousers per hour.

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3.5.3 Track and base. - The cylindrical base for supporting each press group shall be of rigid steel construction and fitted at the bottom with steel wheels in number to adequately support the structure and press group during rotation without undue vibration. The entire structure shall rotate on a circular steel track securely attached to the deck. The design shall be such that roll and pitch of the ship shall not permit shifting of the cylindrical base in excess of 1/4-inch, nor shall it be possible for the cylindrical base to move vertically or for the wheels to slide or lift off the track. For design purposes, a roll of 10 degrees to port and 10 degrees to starboard shall be used.

3.5.4 Steam and air connection. - Steam and air for heating and operation of presses on each cylindrical base shall be supplied from a combination stationary-rotating fitting or fittings located at the center of each component. The fittings shall be so designed that steam and air at required pressures can be transferred from the stationary section to the rotating section without leakage. Provision shall also be provided for adjustment to prevent leakage due to wear. Fittings for steam supply and return, and for air supply shall be provided on the stationary sections for connection of services by others.

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3.5.5 Presses. - Presses to be mounted on the rotating bases are of two types, trouser topping and trouser legging. Each trouser topping press shall be suitable for topping a pair of trousers (one garment) in four lays. Each trouser legging press shall be suitable for legging two pairs of trousers (two garments) in two lays.

3.5.5.1 Each press shall consist of a rugged metal frame supporting the head, buck, work table, and operating mechanism. The head shall be movable and counterbalanced. The yoke shall be fitted with ball, roller, or bronze bearings and shall be arranged so that it will not interfere with placing of garments or with the head making proper contact with the buck when yoke is raised or lowered. Springs, or other suitable means, shall be provided for automatically adjusting the head from end to end and front to back to conform with varying thicknesses of garments being pressed.

3.5.5.2 The head and buck shall be of metal construction, uniformly heated and properly drained. The ironing surface of the head shall be concave in shape, smooth, and either polished or plated. The buck shall be convex in shape to fit the contour of the head and the garments to be pressed. The pressure between head and buck shall be adjustable. The buck shall be fitted with a spring pad with additional padding and covering to insure uniform pressure on garments

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being pressed. The padding and covering shall include a heavy knitted cotton pad, a double-faced canton flannel or table felt, and a tailored cotton covered cloth complete with draw cord and retaining springs.

3.5.5.3 Presses shall be pneumatically operated and shall operate satisfactorily on a working air pressure of 70 p.s.i., minimum and 100 p.s.i., maximum. The head and buck shall be suitable for a working steam pressure of 100 p.s.i.g.

3.5.5.4 The operating mechanism for the presses shall be so arranged that each press head will release or open just prior to reaching the operator's platform, and shall remain open until the press has traveled past the operator's platform, at which time the head shall close and press the garment or garments that have been laid on the buck. A safety shield or flag shall be fitted on each press to prevent the operator from getting his hands caught in the press when it passes the energizing point. The safety shield or flag shall operate just prior to lowering of the press head and shall act to push the operator's hands clear of the press head and buck prior to the head closing. The press head operating mechanism shall be so designed that the press head will not operate to close, or drop due to air pressure failure, under any conditions while in way of the operator's platform. All parts of the press operating mechanism shall be accessible for maintenance purposes.

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3.5.6 Drive mechanism. - Each cylindrical base shall be fitted with an electric motor and drive mechanism to rotate the base and attached presses at the required speed under all conditions of operation without the motor exceeding the permissible temperature rise. The design shall be such that the time for each complete revolution shall not vary more than plus or minus two seconds from the desired speed of rotation when the cylindrical base is level or when it is inclined from a level position to 10 degrees port or starboard. The drive mechanism shall be so designed, or other means provided, that drifting of the cylindrical base, either clockwise or counterclockwise, shall not occur under any condition of operation or ship's motion up to 10 degrees. When de-energized, or in event of a power failure, the cylindrical base shall immediately cease to rotate and remain motionless until the starter switch is activated.

3.5.7 Controls. - Controls for each cylindrical base shall consist of a main motor or power disconnect switch for bulkhead mounting and operational controls located convenient to the operator's position for controlling the machine.

3.5.8 Piping. - All piping, including flexible metallic hose, for supplying steam and air to the presses (and return steam piping) on the upstream side of the rotating union

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for each cylindrical base, shall be supplied by the manufacturer. All piping and flexible metallic hose for steam and air supply shall be suitable for an operating pressure of 100 p.s.i. Pressure gauges, check valves and steam traps shall be suitable for a 100 p.s.i. operating pressure and shall be furnished by the manufacturer.

3.6 Electrical requirements. - All electrical equipment shall conform to the recommended practice for electrical installation aboard ship as specified in IEEE Standard 45. Power supplied to the machine will be 440 volt, 3 phase, 60 cycle, alternating current. The manufacturer shall provide a transformer, transformers or electrical relays, or both, as necessary for other required electrical characteristics. All electrical equipment shall be designed for operation on an ungrounded electrical system but shall operate satisfactorily and without hazard to personnel or equipment with an accidental ground on any conductor. Control equipment shall be splashproof.

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3.6.1 Motor. - Motor shall operate the rotating press unit under all operating conditions without exceeding permissible temperature rise and shall have the following characteristics:

Ambient temperature	50°C.
Service	C.
Enclosure	Totally enclosed.
Cooling	Natural or fan.
Horsepower	As required.
Speed	Constant.
Duty.	Continuous
Winding	Integral horsepower (squirrel-cage induction design B).
Voltage	440 volt, 3 phase, 60 cycle A. C.
Bearings	Ball.
Insulation	Class B, or better.

3.6.2 Controllers. - Contactors shall switch all three lines of three phase circuits and both sides of single phase power circuits. Integral motor power circuits shall be 3 phase and shall be controlled by contactors, which in turn may be actuated by relays. All control circuits shall operate at 115 volts or less. All solenoid valves shall be provided with epoxy encapsulated coils. The control equipment shall have the following characteristics:

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Ambient temperature	50°C.
Enclosure	Splashproof.
Operation	Magnetic.
Type	Across line (A.C.).
Function	Motor starting.
Duty	Continuous.
Protection	Thermal overload and low voltage protection.

3.6.3 Wiring. - Wiring attached to the machine shall be installed in electrical metallic tubing and suitably connected. Wiring on the machine to be connected to devices beyond the machine, such as motor or power disconnect switch, shall terminate in an outlet box or boxes for conduit connection to the devices beyond the machine.

3.7 Technical manuals. - Technical manuals shall be of a commercial type, consisting, in general, of the manufacturer's standard literature illustrations or drawings to concisely outline the proper installation, operation, maintenance, and parts identification of the machine furnished. Each manual shall be suitably bound in a self-cover of durable quality. Two copies shall be submitted to the ordering activity for assignment of a NAVSHIPS number. Two copies of the approved manual shall be packed with each machine. Additional copies

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f the approved manual for stock shall be as specified see 6.1).

3.8 Repair parts. - Repair parts shall accurately fit the machine for which they are specified, and shall be interchangeable with machines of the same type. Repair parts shall be furnished, in general, as follows, or as determined upon completion of Form 4786 (Copies of form 4786 may be obtained from the cognizant Naval Inspector).

3.8.1 Mechanical. - One set of mechanical onboard repair parts shall be furnished with each machine as follows:

1 set of motor bearings.

1 gear of each type and size in drive mechanism.

3.8.2 Electrical. - Electrical onboard repair parts shall consist of one set each, for each size component used, of those items subject to normal wear or breakdown, such as starter coils; solenoid coils; and controller parts subject to wear, such as line contactors.

3.9 Guard rail and securing device. - A suitable and stable guard railing shall be provided for installation around each cylindrical press base and each operator's platform. A device for manually securing each cylindrical press base against rotation, when not in use, shall be provided.

3.10 Identification plate. - An identification plate of durable material shall be secured in a conspicuous place on

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each of the two rotating press components. Marking shall be of a method which will be permanent and legible. Each plate shall include information relative to the manufacturer's name, type, serial number, date of manufacture, contract number, electrical, steam and air characteristics, and the applicable Federal stock number (when given in contract or order).

3.11 Instruction plate. - Each of the two rotating press components shall have an instruction plate made from durable material that is attached to the machine in a location readily visible to the operator. Marking shall be permanent and legible. The instruction plate shall briefly outline operating instructions.

3.12 Workmanship. - The design, materials of construction, fabrication, finish, and operation of the complete rotating trouser pressing machine shall be suitable for the purpose intended and free of all flaws that may affect serviceability and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory

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acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. - Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection and acceptance for the Government will be made after installation aboard the designated ship or ships, unless otherwise specified (see 6.1). The supplier shall supervise the installation of the equipment aboard the designated ship or ships, unless otherwise specified (see 6.1). Final acceptance inspection shall determine compliance with all requirements of this specification and shall include the functional test specified in 4.4.

4.2 Preproduction. - Prior to the start of production and within 60 days after the date of award of the contract or order, the complete automatic rotating trouser topping press component of one complete unit shall be subjected to the tests specified in 4.3.1 through 4.3.4.. These tests shall be witnessed by an authorized Government representative and

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production of the remaining equipments under the contract or order is contingent upon the preproduction component satisfactorily passing all tests.

4.3 Tests. - A platform shall be constructed of a size to adequately support the preproduction rotating trouser topping press component during tests. The platform shall be so constructed that it can be raised at each side and at each end a distance equivalent to 10 degrees. Only one side or one end is to be raised at any one time. Tests shall be started with the test component secured to the platform and the platform in the level position.

4.3.1 Power input. - The test component shall be connected to an appropriate source of electric power and the power input measured over a one hour operating period.

4.3.2 Insulation. - The insulation resistance of power carrying electrical components shall be measured with a 500 volt, d.c., insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

4.3.3 Pressure test. - Each press head and buck, and all steam and air piping shall be subjected to test at 150 percent of the specified working pressures for not less than one minute to determine that there are no leaks or damage to the equipment.

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4.3.4 Inclination test. - The test component shall be inclined 10 degrees, by raising the test platform at one end or one side, in four different positions of 90 degree increments. The test component shall be operated as follows in each of the four positions to determine compliance with the requirements of 3.5.5.4 and 3.5.6:

(a) Operate the component for ten complete revolutions to determine that the machine conforms to design speed, that the machine rotates at a steady speed (does not alternately run slow and fast due to angle of inclination), that presses open and close at proper intervals, and that safety flags or shields will accomplish the purpose for which intended.

(b) Stop the machine as each press reaches the center of the operator's platform, close off air to simulate an air pressure failure, and determine that the press head remains in the open position.

(c) Alternately start and stop the machine every five seconds for a period of not less than three minutes to determine that the machine stops immediately when de-energized and that the machine does not drift forward or backward.

4.4 Functional test. - After installation aboard ship, the complete unit shall be operated to demonstrate that 200 trousers per hour can be finished in accordance with 3.5.2.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.

5.1.1 Pressing equipment, trouser, automatic, rotating, and detached parts. -

5.1.1.1 Preservation and packaging. - Each automatic trouser pressing unit and detached parts shall be preserved and packaged in accordance with the supplier's commercial practice to provide protection from contamination and physical damage in general handling, storage and issue.

5.1.1.2 Packing. - Each automatic trouser pressing unit and detached parts shall be packed in containers in such a manner as to insure acceptance by common carrier and safe delivery at destination. Containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include name of item; Federal stock number (if provided in contract or order); contract or order number; contractor's name; and the destination.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Voltage for motors if other than specified (see 3.5.1).

(c) Quantity of manuals for stock (see 3.6).

(d) Repair parts, if other than specified (See 3.7).

(e) If final inspection and acceptance are to be at destination other than specified (see 4.1.1).

(f) If installation of equipment is not to be supervised by supplier (see 4.1.1).

(g) Preservation, packaging, packing or marking requirements if other than specified (see 5.1).

6.2 Preproduction. -

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 3510-N035Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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